

Problem 1 (60 Points): Consider a Gaussian Mixture Model (GMM) with two well-separated modes:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 1) + 0.5 \cdot \mathcal{N}(\theta; 3, 1),$$

where $\mathcal{N}(\theta; \mu, \sigma^2)$ is a Gaussian density with mean μ and variance σ^2 . Our goal is to approximate the above GMM using the variational distribution:

$$q_\lambda(\theta) = \mathcal{N}(\theta; \mu, \sigma^2),$$

where the variational parameter $\lambda = (\mu, \sigma)$.

- i) (20 Points) Derive the explicit update rule for both μ and σ using the re-parameterization trick.
- ii) (20 Points) Write a computer program to find the optimal variational distribution for the above GMM model. Try multiple initial values for μ and σ and describe how the value of μ and σ change over the iterations.
- iii) (20 Points) Repeat Part ii) for the following updated GMM model:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 4) + 0.5 \cdot \mathcal{N}(\theta; 3, 0.25).$$